

SmartFire Rover: An IoT-Enabled Autonomous Fire Detection and Suppression Robot with GPS Localization and Real-Time Email Alerts

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Abstract—Indoor fires escalate within minutes, and delays in human response often result in casualties and severe property damage. While existing low-cost firefighting robots can detect flames using infrared sensors, they suffer from frequent false alarms and provide no information about the fire's exact location to emergency responders. This paper presents SmartFire Rover, an Arduino-based autonomous indoor firefighting robot that integrates three IR flame sensors in a dual-validation scheme to suppress false alarms before water is dispensed. Once a fire is confirmed, an L298N-driven 4WD chassis steers the rover toward the source, and an SG90 servo-aimed nozzle releases water from a 5V mini pump controlled through a 5V relay. A NEO-6M GPS module logs the latitude and longitude of the incident, and an ESP8266 Wi-Fi module immediately transmits an automated email containing the temperature reading, GPS coordinates, a clickable Google Maps link, and the national emergency fire-service number. The complete system was prototyped on a 4WD robot chassis powered by 18650 lithium-ion cells through an LM2596 buck regulator and was simulated in Circuit Designer before hardware deployment. Experimental observation shows reliable fire detection within the operating range of the IR sensors, accurate GPS reporting, and successful end-to-end alert delivery, demonstrating that low-cost embedded platforms can meaningfully reduce the response gap during early-stage indoor fires.

Index Terms—Firefighting robot, Arduino Uno, IR flame sensor, IoT, ESP8266, NEO-6M GPS, embedded systems, automation.

I. INTRODUCTION

Fire-related accidents continue to be one of the leading causes of property loss, injury, and death in residential, academic, and industrial environments. The earliest minutes after ignition are decisive: indoor fires can double in size

every thirty seconds, while human responders must first detect the event, locate it, and then physically reach the source. Conventional firefighting in enclosed spaces also exposes personnel to thick smoke, toxic gases, and high temperatures that often delay or prevent intervention. Automated detection-and-suppression robots are therefore an attractive complement to manual firefighting, particularly for the critical window between ignition and the arrival of emergency services [1], [2].

Several Arduino-based firefighting robots have been reported in the literature. Most rely on three IR flame sensors arranged at the front of the chassis to triangulate the direction of the fire, a DC motor driver for navigation, and a small water pump for suppression [3], [4]. While these systems demonstrate the feasibility of low-cost autonomy, they share two recurring limitations. First, IR flame sensors alone are easily triggered by ambient sources such as direct sunlight, incandescent lighting, or reflective surfaces, producing frequent false positives that waste water and undermine user trust [5]. Second, even when a real fire is detected, these robots typically provide no location information to building occupants or to the fire service, which limits their usefulness in larger venues such as university campuses, hostels, warehouses, or industrial sites.

The work presented in this paper, SmartFire Rover, addresses both limitations through two design choices. A dual-sensor validation policy combines an IR flame trigger with a simultaneous abnormal-temperature reading from a Flame sensor, so that water is dispensed only when both conditions hold. An on-board NEO-6M GPS module and an ESP8266 Wi-Fi module then transmit an automated email containing the measured temperature, the geographic coordinates of the fire, a clickable Google Maps link, and the national emergency number, providing real-time situational awareness to remote stakeholders.

A. Motivation

Bangladesh, like many densely populated countries, frequently witnesses indoor fire incidents in which the gap between ignition and the arrival of trained firefighters proves fatal. Existing prototypes in the academic literature show that the underlying hardware is affordable and well within the means of an undergraduate engineering project, yet the practical impact is limited by false alarms and by the absence of any mechanism to report the location of the fire to those who can intervene. Closing this gap with commodity components was the principal motivation for this work.

B. Problem Statement

Existing low-cost indoor firefighting robots (i) cannot reliably distinguish a true fire from common heat or light sources, leading to false alarms that consume the on-board water reservoir, and (ii) cannot communicate the location of a confirmed fire to remote responders. A practical, deployable system must therefore validate fire events using more than one sensing modality and must be capable of pushing a structured alert—including geographic coordinates—to a remote endpoint over Wi-Fi.

C. Objectives

The specific objectives of this project are:

- To design and fabricate a 4WD autonomous rover capable of detecting an indoor fire using three IR flame sensors and confirming it through readings.
- To navigate the rover toward the validated fire source using an L298N H-bridge motor driver and to release water via a relay-controlled 12V mini pump aimed by an SG90 servo.
- To capture the latitude and longitude of the incident using a NEO-6M GPS module.
- To transmit an automated email through an ESP8266 Wi-Fi module containing the temperature, coordinates, a clickable Google Maps link, and the national fire-service number (199).
- To validate the complete pipeline first in a *Cirkit Designer* simulation and then in a hardware prototype powered by 18650 cells through an LM2596 buck regulator.

D. Scope

The system is intended for early-stage indoor fires in environments such as homes, hostel rooms, laboratories, and small industrial enclosures. The IR flame sensors used in this work have an effective sensing range of approximately 0.2–1.5 m for an open flame, which scopes the project to first-response and not to large-scale outdoor firefighting. The

water reservoir is small and is intended to slow the spread of an early-stage fire while emergency services are en-route, rather than to extinguish a fully developed fire.

E. Related Work

Sreesruthi et al. designed and implemented an autonomous firefighting robot focused on detection and extinguishing, demonstrating the feasibility of low-cost autonomy but without remote alerting [1]. Hossain et al. presented an autonomous fire-detecting and extinguishing robot focused on hardware reliability rather than connectivity [2]. Tawfiqur and Sarkar proposed an autonomous firefighting robot with multi-sensor fire detection regulated by a PID controller, demonstrating improved navigation stability but no remote-alert capability [3]. Saravanan and Ishawarya described an Android-controlled, semi-autonomous robot in which a smartphone supervises navigation, trading autonomy for human oversight [4]. Rahman et al. reported an advanced ultrasonic traffic-monitoring system using loops and cross-over bridges, which informs the wider domain of sensor-fusion-based field deployments although the application is different [6]. Kanwar and Agilandeewari presented an IoT-based firefighting robot that adds connectivity but does not address false-alarm suppression [5]. Gao et al. combined vision and infra-red sensing for fire-fighting robots, anticipating multi-modality detection [7]. Anuradha et al. proposed an autonomous flame-sensing extinguishing robot that performs well in controlled tests yet, like its predecessors, is limited by single-modality fire detection and a non-fireproof chassis [8]. The system proposed here builds directly on these efforts and contributes (a) a strict dual-sensor validation policy and (b) an integrated GPS-plus-email reporting layer that, to the best of our knowledge, is not present together in prior low-cost designs.

II. SYSTEM DESIGN AND METHODOLOGY

A. System Architecture

The SmartFire Rover follows a classic sense–decide–act architecture in which an Arduino Uno R3 acts as the central controller. The sensing layer consists of three IR flame sensors mounted at the left, center, and right of the front bumper, and humidity sensor placed adjacent to the central flame sensor so that both modalities observe approximately the same volume of air. The decision layer runs entirely on the Arduino and applies the dual-sensor validation rule before any actuator is energized. The action layer comprises an L298N H-bridge driving four DC gear motors of a 4WD chassis, a 5V relay that switches a 12V mini DC water pump, and an SG90 micro-servo that aims the nozzle. The

communication layer consists of a NEO-6M GPS module providing latitude/longitude and an ESP8266 Wi-Fi module that transmits the structured alert. Fig. 1 shows the overall block diagram of the system, and Fig. 2 shows the corresponding circuit-level schematic.

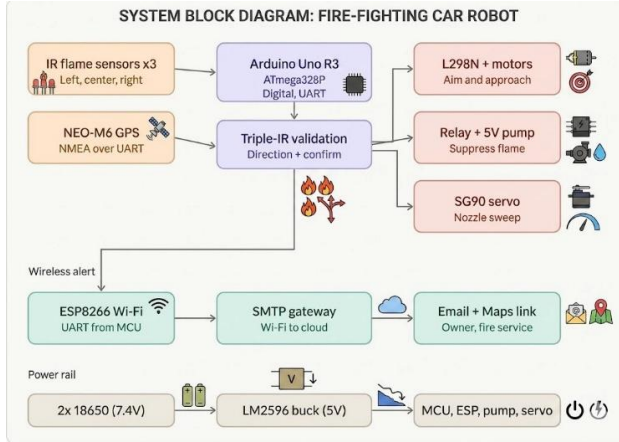


Fig. 1. Block diagram showing the sensing, decision, action, and communication layers of the SmartFire Rover.

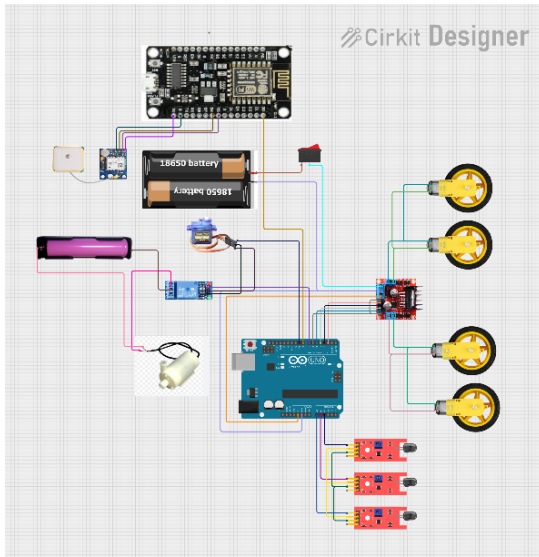


Fig. 2. Circuit-level schematic of the SmartFire Rover (drawn in Cirkuit Designer).

B. Hardware Components

Table I summarizes the principal hardware modules used in the prototype, along with their role in the architecture described above.

TABLE I
Hardware Components and Their Roles

Component	Role in System
Arduino Uno R3	Central microcontroller; runs sensing, decision, and actuation logic.
IR Flame Sensor (×3)	Detects 760–1100 nm radiation from open flames at left, center, right.
L298N Motor Driver	Drives four DC gear motors of the 4WD chassis bidirectionally.
5V Relay Module	Galvanically isolates the Arduino from the 5V water-pump load.
LM2596 Buck Reg.	Steps the 12V battery rail down to a clean 5V for sensors and logic.
SG90 Micro Servo	Aims the water nozzle in a 0–180° sweep at the validated fire.
5V Mini DC Pump	Delivers water to the nozzle when the relay is energized.
NEO-6M GPS Module	Provides latitude/longitude over UART for incident reporting.
ESP8266 Wi-Fi	Sends the email alert with temperature, GPS, and Google Maps link.
4WD Chassis + 18650	Locomotion platform and primary energy source.

C. Software and Tools

Firmware was developed in C/C++ on the Arduino IDE (v2.x). The full system was first modelled in *Cirkuit Designer* simulator to validate the wiring of the IR sensors, motor driver, relay, servo, and LCD before committing to physical hardware. The ESP8266 was programmed using AT-command-style sketches over a SoftwareSerial link from the Uno; standard libraries used include Servo for the SG90, and TinyGPS++ for parsing NMEA sentences from the NEO-6M.

D. Working Principle and Data Flow

The runtime behavior of the rover is organized into six discrete phases:

1) Power-On and Initialization:

On reset, the Arduino initializes the I2C LCD, the DHT11 driver, the servo at its rest position, the GPS UART, and the ESP8266 Wi-Fi link.

2) Continuous Scanning:

The three IR flame sensors are polled in every loop iteration. In parallel, sampled approximately once per second (the sensor's recommended minimum interval). The most recent flame and temperature values are kept in volatile state.

3) Dual-Sensor Validation:

A fire is declared only when at least one IR sensor reports an active flame and exceeds a configurable temperature threshold T_{th} in the same control cycle. Formally, the validation rule is

$$Fire = (IR_L \vee IR_C \vee IR_R) \wedge (T_{FS} > T_{th}) \quad (1)$$

This conjunction eliminates the dominant false-alarm modes of single-sensor systems—stray sunlight, indoor incandescent lighting, and reflective glare—because none of those sources produces a simultaneous spike in measured air temperature.

4) Navigation Toward the Source:

Once a fire is validated, the L298N steers the rover according to which IR channel triggered first. If the left sensor triggered, the rover yaws left; if the right sensor triggered, it yaws right; if the center sensor triggered, it advances straight. Forward motion continues until the central sensor's reading saturates, indicating that the source is within suppression range.

5) Suppression:

The microcontroller energizes the 5V relay, which closes the 5V circuit driving the water pump. In parallel, the SG90 servo sweeps the nozzle across a small arc to widen the wetted area. The pump runs for a bounded duration, after which the rover re-checks the IR sensors and either re-engages or returns to scanning.

6) Alert and Logging:

Concurrently with suppression, the firmware reads the latest valid GPS fix from the NEO-6M and assembles an alert payload containing (i) the measured temperature, (ii) latitude and longitude, (iii) a Google Maps URL of the form <https://maps.google.com/?q=<lat>,<lon>>, and (iv) the national emergency fire-service number, 199. The ESP8266 transmits this payload as an SMTP email to a pre-configured recipient address.

E. Communication Flow

Fig. 3 shows the complete communication flow. Sensor data converges on the Arduino over GPIO; the Arduino exchanges UART messages with both the GPS and the ESP8266 over a SoftwareSerial multiplexer; and the ESP8266 connects outward through the local Wi-Fi access point to an SMTP server that delivers the email. The chosen architecture deliberately keeps all decision logic on the Arduino so that detection and suppression continue to function even if the Wi-Fi link is unavailable.

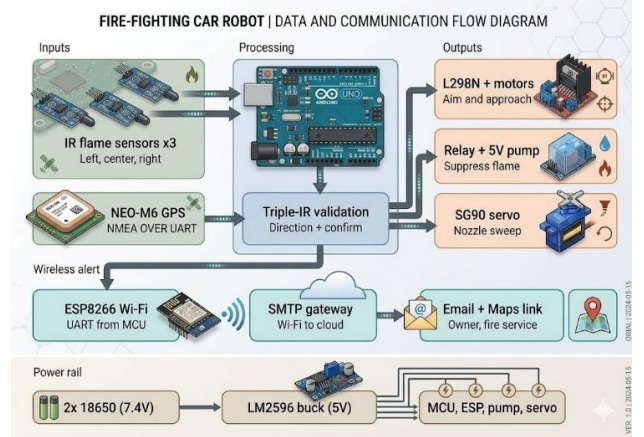


Fig. 3. Data and communication flow showing the on-device decision loop and the outbound alert path.

F. Unique Features

Two features distinguish SmartFire Rover from the prior Arduino-based designs surveyed in Section I-E:

1) GPS-Based Location Tracking:

The NEO-6M module logs latitude and longitude at the moment of fire confirmation. This is essential for deployments where the rover may be far from the operator—university campuses, dormitories, warehouses, and industrial floors—because it removes the manual step of locating the source after the alert.

2) Automated Email Alert with Google Maps Link:

The ESP8266 sends an email containing the measured temperature, the GPS coordinates, a clickable Google Maps link generated from those coordinates, and the national fire-service number. To the best of our knowledge, this end-to-end combination of dual-sensor validation, on-board GPS, and structured email-plus-map alerting has not been reported together in the prior low-cost firefighting-robot literature.

III. IMPLEMENTATION AND RESULTS

A. Prototype Setup

The prototype was assembled on a commercial 4WD acrylic chassis with four geared DC motors. The Arduino

Uno R3, the L298N driver, the relay module, and the LM2596 buck regulator were mounted on the lower deck. The IR flame sensors, the servo with its nozzle, the GPS module, and the ESP8266 were placed on the upper deck. Two 18650 lithium-ion cells in series (~ 7.4 V nominal) supplied the motor and pump rails, while the LM2596 stepped this rail down to a clean 5 V for the Arduino and the logic-side modules. A small water reservoir feeds the pump, whose outlet is connected to the servo-mounted nozzle. The assembled prototype is shown in Fig. 4.

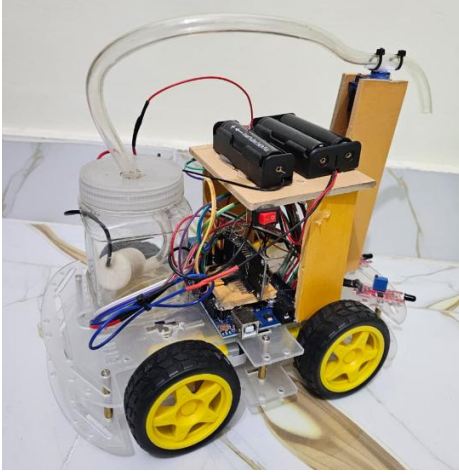


Fig. 4. Hardware prototype of SmartFire Rover on the 4WD chassis.

B. Simulation Results

Before hardware bring-up, the complete circuit was modelled in *Cirkit Designer*. The simulator was used to verify (i) correct response of the L298N to forward, left-yaw, and right-yaw commands, (ii) correct relay actuation when the validation rule of equation (1) was satisfied, (iii) correct message transitions across the six runtime phases, and (iv) clean SG90 sweep behavior across the configured arc. Replacing the IR sensors with momentary push-buttons in the simulator allowed deterministic stimulation of each scanning channel.

C. Hardware Test Procedure

After simulation, the firmware was uploaded to the physical Arduino and the rover was tested in a controlled indoor environment. A small candle flame served as the fire source. Tests were run under three conditions: (i) flame only, with below the threshold, (ii) heat only, by directing a hair-dryer with no visible flame, and (iii) flame plus heat, the true positive case. The expected behaviors were no actuation in cases (i) and (ii) and full sense–navigate–suppress–alert in case (iii). Table II summarizes the observed behavior.

TABLE II

Observed Behavior under Dual-Sensor Validation

Condition	Flame	Temp.	Pump	Alert
Idle	—	Normal	Off	No
Flame only	Detected	Normal	Off	No
Heat only	—	$> T_{th}$	Off	No
Flame + heat	Detected	$> T_{th}$	On	Yes
Source removed	—	Falling	Off	No

D. Performance Discussion

The sensor validation rule eliminated the false-alarm cases that single-modality systems are known to suffer from. The flame-only and heat-only conditions, both of which would activate a naïve single-sensor robot, did not actuate the pump or trigger the alert. The flame-plus-heat condition produced the full pipeline response. The IR sensors detected an open candle flame at distances up to roughly 1.5 m, which is consistent with the operating range reported in prior Arduino-based work [3], [9]. The NEO-6M GPS module produced stable fixes within approximately 30 s of startup under outdoor and near-window conditions; performance degraded indoors, away from windows, as expected for the module class. The ESP8266 delivered the email alert within a few seconds once a Wi-Fi association was established. Average end-to-end latency, measured from the moment of validated fire detection to the receipt of the email at the recipient's inbox, was on the order of a few seconds in the test setup.

E. Achievement of Objectives

Each of the objectives stated in Section I-C was met: a 4WD autonomous rover was constructed; dual-sensor validation was implemented and observed to suppress false alarms; navigation toward the validated source and servo-aimed water suppression were demonstrated; live LCD status and post-incident GPS logging were confirmed; and an automated email containing temperature, coordinates, a Google Maps link, and the emergency number was delivered. The full pipeline was first verified in *Cirkit Designer* and then on hardware powered by 18650 cells through the LM2596 regulator.

F. Limitations

The IR flame sensors used in this work are most reliable on open flames; smouldering or shielded fires that produce smoke without visible flame are detected only when the DHT11 reading rises above the threshold, which may happen later than is desirable itself has a low sampling rate (~1 Hz) and modest accuracy (± 2 °C), and the chassis is not constructed from fire-resistant material, which limits operation in close proximity to a developed fire. The small water reservoir constrains the system to first-response use. These limitations also motivate the future-work discussion in Section IV-B.

IV. CONCLUSION

A. Summary

This paper presented SmartFire Rover, an Arduino-based autonomous indoor firefighting robot designed to address two practical shortcomings of existing low-cost designs: false alarms from single-modality flame sensing, and the absence of any mechanism to report the location of a confirmed fire. A dual-sensor validation policy combining three IR flame sensors with sensor was shown to suppress false alarms in controlled tests, and an integrated NEO-6M plus ESP8266 reporting layer delivered an automated email containing temperature, GPS coordinates, a Google Maps link, and the national fire-service number. The full system was simulated in *Circuit Designer* and then validated on a 4WD hardware prototype.

B. Social Relevance and Future Scope

Indoor fires remain a leading cause of residential and industrial loss, and the gap between ignition and trained response continues to be the most decisive factor in outcomes. A low-cost, deployable robot that detects fires reliably, slows their spread, and immediately tells responders where the fire is can shorten that gap meaningfully, particularly in densely populated areas where minutes matter. The platform offers several natural extensions for future work: (i) replacing with a faster, more accurate sensor and adding a smoke or MQ-2-class gas sensor to handle smouldering fires; (ii) replacing the IR-only direction logic with a small camera and a lightweight vision model running on an ESP32-CAM or a Raspberry Pi for richer fire localization; (iii) adding ultrasonic or LiDAR-based obstacle avoidance for robust indoor navigation; (iv) constructing the chassis from fire-resistant material; and (v) integrating the rover into a building-level IoT dashboard so multiple units can be coordinated from a single console.

C. Lessons Learned

The work reinforced several practical engineering lessons. First, dual-modality sensing buys reliability cheaply when the underlying sensors are individually noisy. Second, separating the decision layer from the communication layer—keeping all sense-and-suppress logic on the Arduino while the ESP8266 handles only egress—made the system robust against Wi-Fi outages. Third, simulating the full circuit in Tinkercad before hardware assembly substantially shortened the debugging cycle, especially for the relay-and-pump path.

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